

Blind Hole Thread Depth Charts

Use these charts to model full thread depth and tap-drill depth to the full-diameter shoulder. Metric depths are shown in inches and millimeters; UNC depths are in inches. Metric sizes use coarse pitch.

Inch full thread depth = $1.5 \times \text{diameter (in)} + .125''$

Metric full thread depth = $1.5 \times \text{diameter (mm)} + 3.25 \text{ mm}$

Extra drill depth = greater of $.120''$ (3.048 mm) or $4 \times \text{pitch}$

Drill depth to shoulder = full thread depth + extra drill depth

Metric Coarse Threads M3 to M20

Thread callouts give nominal diameter $\times$ pitch in millimeters.

Thread size	Full thread depth (in)	Full thread depth (mm)	Drill shoulder depth (in)	Drill shoulder depth (mm)
M3 $\times$.5	0.31	7.8	0.43	10.9
M3.5 $\times$.6	0.34	8.5	0.46	11.6
M4 $\times$.7	0.37	9.3	0.49	12.4
M5 $\times$.8	0.43	10.8	0.56	14.0
M6 $\times$ 1	0.49	12.3	0.65	16.3
M7 $\times$ 1	0.55	13.8	0.71	17.8
M8 $\times$ 1.25	0.61	15.3	0.81	20.3
M10 $\times$ 1.5	0.72	18.3	0.96	24.3
M12 $\times$ 1.75	0.84	21.3	1.12	28.3
M14 $\times$ 2	0.96	24.3	1.28	32.3
M16 $\times$ 2	1.08	27.3	1.40	35.3
M18 $\times$ 2.5	1.20	30.3	1.60	40.3
M20 $\times$ 2.5	1.31	33.3	1.71	43.3

Rounding: round full thread depth and extra drill depth upward to $.01''$ or $.1 \text{ mm}$, then add for drill depth. Metric inch and millimeter columns are calculated independently; use one unit system consistently. This preserves tap clearance after rounding.

Inch UNC Threads Number 8 to 1 Inch

All depth values are in inches. Pitch = $1 \div$ threads per inch.

Thread size	Full thread depth (in)	Extra drill depth (in)	Drill depth to shoulder (in)
#8-32	0.38	0.13	0.51
#10-24	0.41	0.17	0.58
#12-24	0.45	0.17	0.62
1/4-20	0.50	0.20	0.70
5/16-18	0.60	0.23	0.83
3/8-16	0.69	0.25	0.94
7/16-14	0.79	0.29	1.08
1/2-13	0.88	0.31	1.19
9/16-12	0.97	0.34	1.31
5/8-11	1.07	0.37	1.44
3/4-10	1.25	0.40	1.65
7/8-9	1.44	0.45	1.89
1-8	1.63	0.50	2.13

Application Notes

Tooling: the 4-pitch allowance assumes a blind-hole tap with a cutting lead no longer than 3 pitches, plus 1 pitch of clearance. Confirm actual tap geometry, any projecting point, chip space and process requirements before adopting this rule. Longer-lead taps can require more depth.

Drill point: these depths end at the shoulder. The drill point extends farther; include it when checking remaining wall thickness and nearby passages. Fine threads require recalculation using their actual pitch.